UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG

Human Research Ethics Committee: (Medical)
FWA Registered No IRB 00001223

SECRETARIAT: Suite 189, Private Bag x2600, Houghton 2041, South Africa Tel: +27-11-274 9200

Ms D Lakhoo

EMAILED

10 November 2025

Wits RHI - Esselen Street
Hillbrow Health Precinct
22 Esselen Street
Hillbrow
2001

Email: dlakhoo@wrhi.ac.za

Dear Ms Lakhoo,

PROTOCOL: HEAT002 - INNOVATIVE MACHINE LEARNING AND MULTI-SOURCE DATA ANALYSIS TOWARDS DEVELOPMENT OF AN URBAN HEAT-HEALTH EARLY WARNING SYSTEM FOR AFRICAN CITIES

ETHICS REFERENCE NO: 220606

RE : APPROVAL FOR PROTOCOL VERSION 3.0 DATED 09 OCTOBER 2025

We acknowledge receipt of your letter dated 28 October 2025 with the following documentation pertaining to the above-captioned study.

Amendment Date:	09-Oct-25	Amendment Version:	Protocol Version 3.0
Amendment Number:		Received Date:	28-Oct-25

The following has been reviewed and approved by the Initial Reviewers of the University of the Witwatersrand, Human Research Ethics Committee: (Medical)

* HEAT002 Protocol Version 3.0 dated 09 October 2025
* HEAT002 Letter of Amendment #2

Ethics Approval Date: 10 November 2025

The University of the Witwatersrand, Human Research Ethics Committee (Medical) Approval Granted for the above mentioned study is valid for five years. Where required by Sponsor to have approval on a more frequent basis it remains the responsibility of the Sponsor and Investigator to apply for continuing review and approval, or for the duration of the study.

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1. THIS APPROVAL IS SUBJECT TO THE FOLLOWING PROVISOS:

* A copy of the SAHPRA Approval and/or SAHPRA Notification letter must be submitted to the Ethics Regulatory Office Secretariat before the study commences / or where an Amendment may be implemented (IF SAHPRA APPROVAL / NOTIFICATION IS APPLICABLE). It remains the responsibility of the Principal Investigator and/or Sponsor to ensure that the relevant approvals are in place.

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* The study is conducted according to the protocol submitted to the University of the Witwatersrand, Human Research Ethics Committee (Medical). As a result, any amendments to the protocol must be submitted for approval, to the Human Research Ethics Committee (Medical).

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* During the study, the University of the Witwatersrand, Human Research Ethics Committee (Medical) is informed immediately of :

- Any Serious Adverse Events or Unexpected Adverse Drug Reactions, which, in the Investigator and/or the Sponsor's opinion are suspected to be related to the study. This includes study IP, concomitant or comparator products and study investigations. (Refer to POL-HREC–001 and SOP-HREC–005, Item 4).
- Any data received during the study which, may cast doubt on the safety and/or validity of the continuation of the study .

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* The University of the Witwatersrand, Human Research Ethics Committee (Medical) is notified of any decision to discontinue the study and the reason stated.

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* The Investigators are authorised by this approval to participate in this study. Additional Investigators shall be submitted to the University of the Witwatersrand, Human Research Ethics Committee (Medical) for approval prior to their participation in the study.

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* In the event of an authorised Investigator ceasing to participate in the study, the University of the Witwatersrand, Human Research Ethics Committee (Medical) must be informed and the reason for such cessation given.

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2. PRINCIPLES OF INFORMED CONSENT:

* The University of the Witwatersrand, Human Research Ethics Committee (Medical) requires that in all studies, the Principles of Informed Consent/Assent as per SA GCP 2020 are adhered to. This applies to volunteers as well as patients.

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3. PROGRESS REPORTS:

* The University of the Witwatersrand, Human Research Ethics Committee (Medical) requests that the SAHPRA Progress Reports be submitted either: six-monthly from the date of the SAHPRA approval, six-monthly from the date of Wits HREC (Medical) approval, where SAHPRA approval is not applicable, or in March and September, to the HREC Secretariat Office - email: EthicsRegulatory@witshealth.co.za and a report of the final results after the conclusion of the study. More frequent monitoring may be requested in high/very high risk studies e.g. First-In-Human (FIH) studies (if applicable).

4. PARTICIPANT REIMBURSEMENT/REMUNERATION FOR TIME, INCONVENIENCE AND EXPENSE (TIE):

* The University of the Witwatersrand, Human Research Ethics Committee: (Medical) is in agreement that reimbursement per visit is according to the South African Health Products Regulatory Authorities TIE Principles (Time, Inconvenience and Expense). Please note that this provision of funds is not in any way a payment nor an incentive, but only to cover participants TIE. Time is reimbursed based on the national minimum wage of R28.79 per hour.

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5. TRANSPORT AND STORAGE OF HUMAN BIOSPECIMEN OUTSIDE SOUTH AFRICA (AS PER SOP):

* If human biospecimens are to be stored for future analysis, and it is planned that such analysis is done outside South Africa, it is recommended that the biospecimens are also stored at a facility in SA. The HREC must approve the storage at this facility and the release of samples, together with the Material Transfer Agreement to facilities outside the country. The approval must be protocol specific. Where storage is intended to be at a facility outside SA, motivation for such storage, together with comprehensive information on the storage facility and the Material Transfer Agreement must be submitted.

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6. STORAGE OF BIOSPECIMENS/DATA FOR FUTURE RESEARCH (INCLUDING GENETIC TESTING)

* The Wits Human Research Ethics Committee: (Medical) must approve future research which is not related to the original study.

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7. GOOD CLINICAL PRACTICE

SAHPRA requires Good Clinical Practice (GCP) Training for all Investigators and that GCP training be renewed every three (3) years.
Initial GCP training must be done in the form of a live facilitator-led course which may be in-person or virtual.
GCP Refresher courses may, however, be done online.
The GCP training must be based on the latest South African Good Clinical Practice: Clinical Trial Guidelines. Third Edition (SA GCP 2020).

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The above has been noted for the Ethics Committee information and records.

KINDLY FORWARD TO THE RELEVANT INVESTIGATORS / CRA / STUDY CO-ORDINATORS

Regards,

Paul Ruff

PROF PAUL RUFF

For and on behalf of the Human Research Ethics Committee: (Medical)

Processed by: Thashin Reddy - Tel: 011 274 3353

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Document 1 of 3

Document Name: 220606 - Approval for Protocol Version 3.0 dated 09 Oct 2025.pdf

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Signers

Paul Ruff

Paul Ruff

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CC Recipients

Jennifer jennifer.palmer@witshealth.co.za

Supporting documentation

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